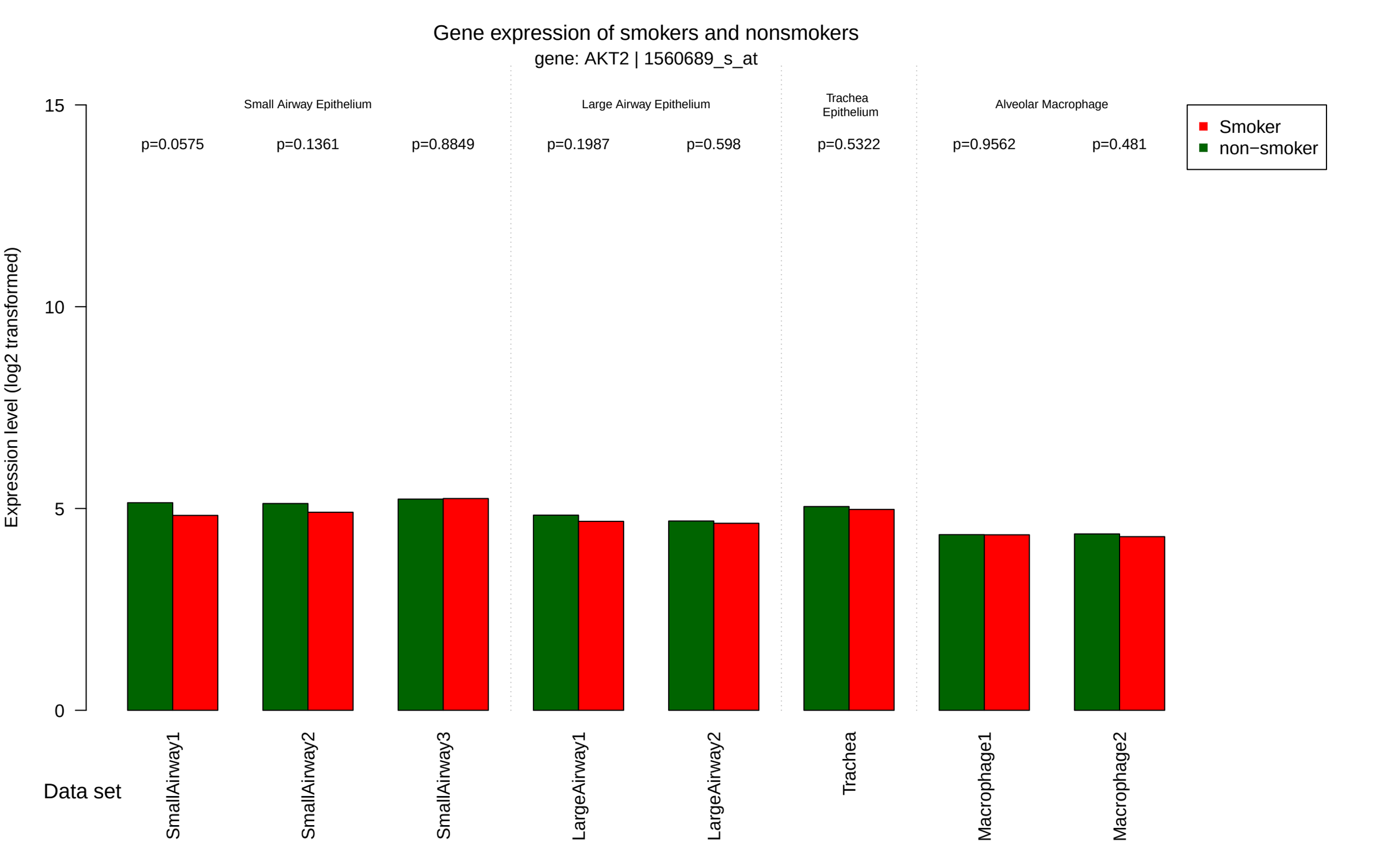
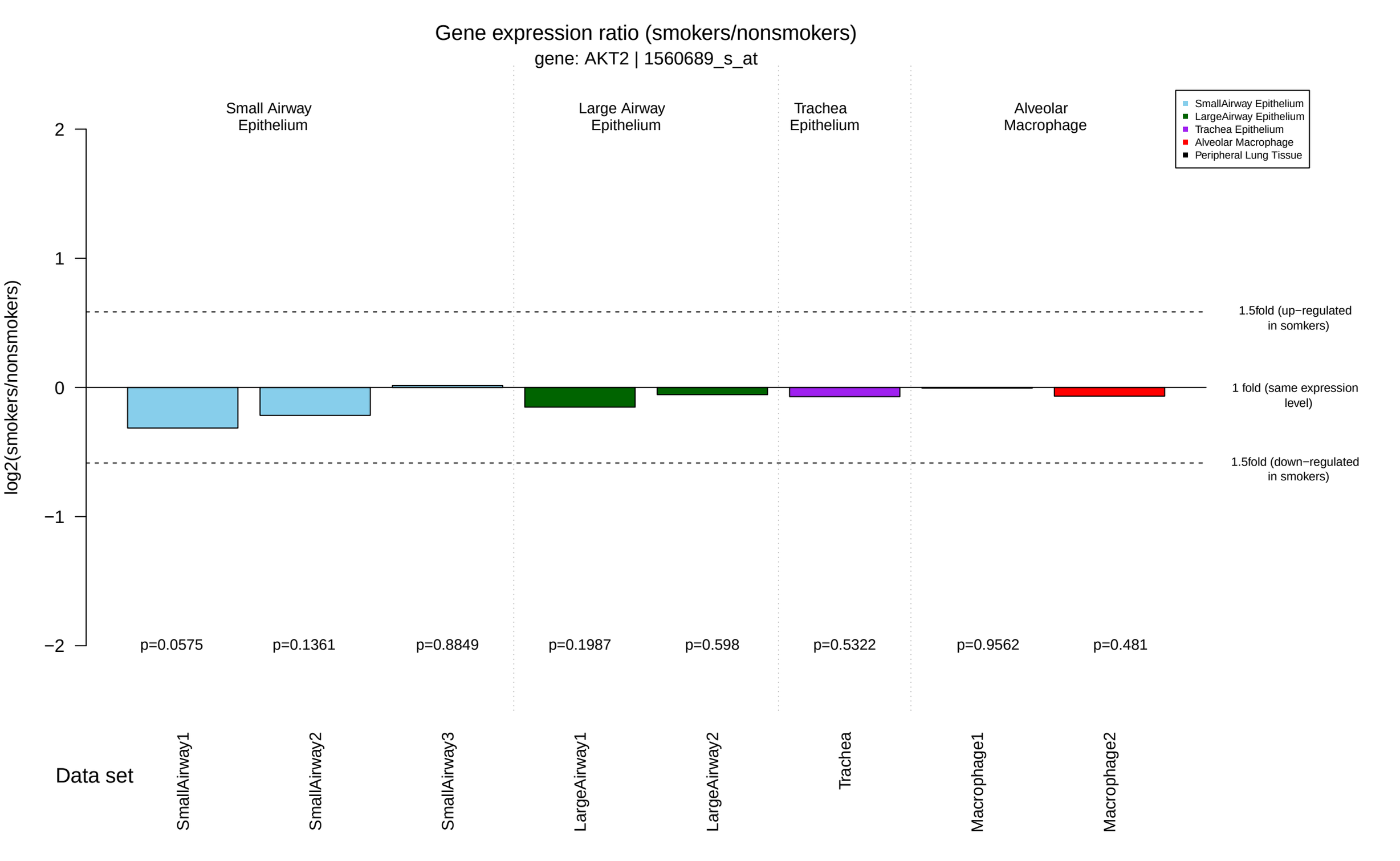
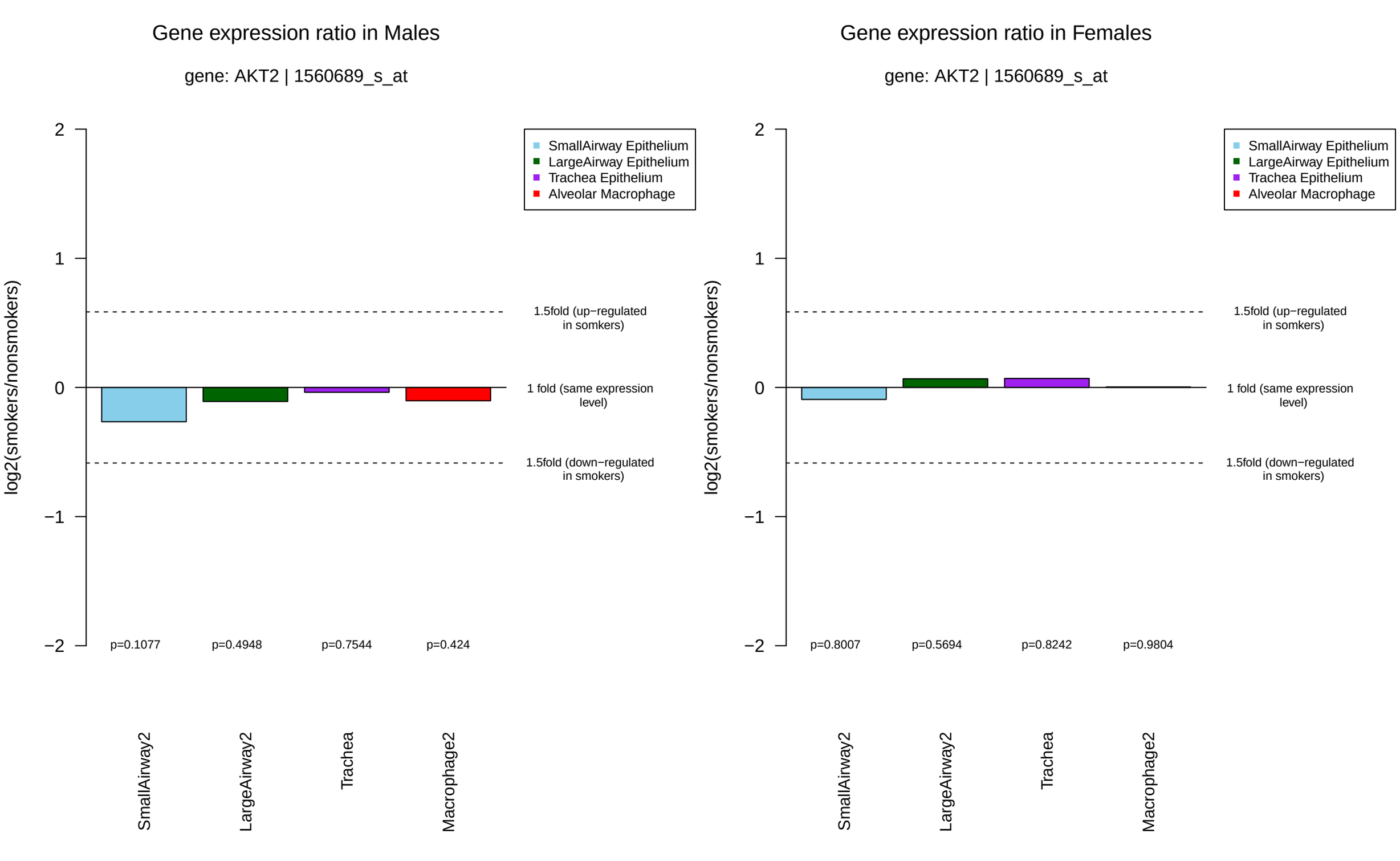
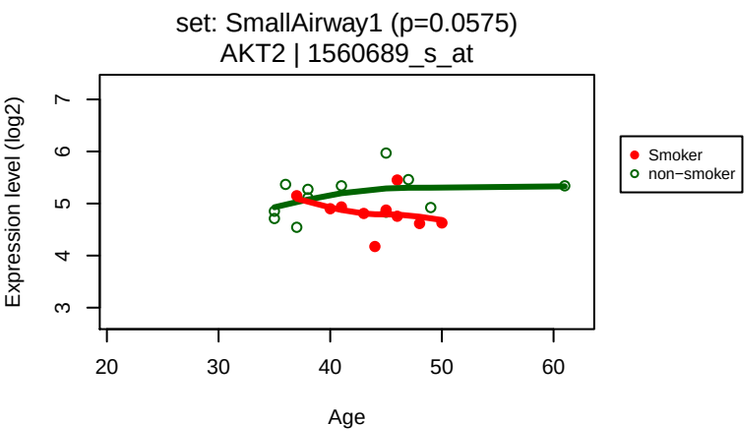
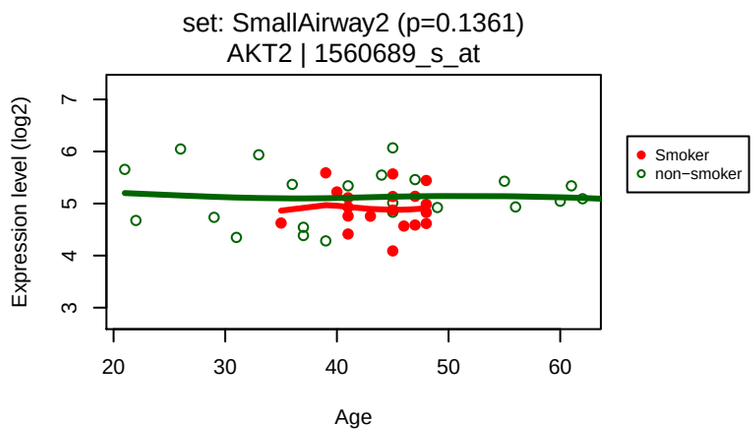
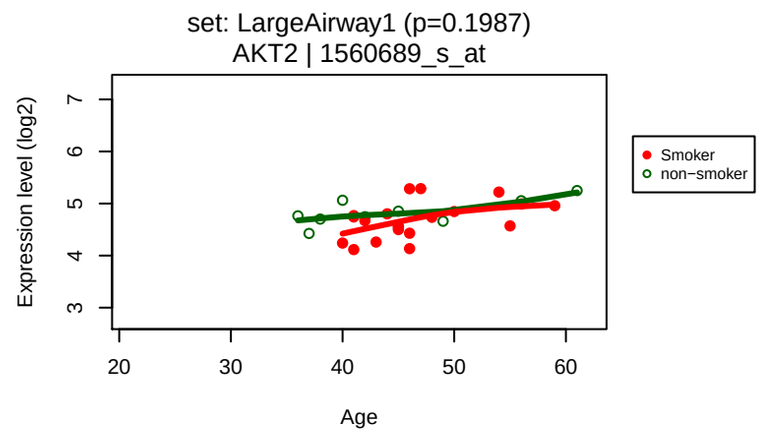

Gene expression of smokers and nonsmokers
gene: AKT2 | 1560689_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0575
p=0.1361
p=0.8849
p=0.1987
p=0.598
p=0.5322
p=0.9562
p=0.481
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

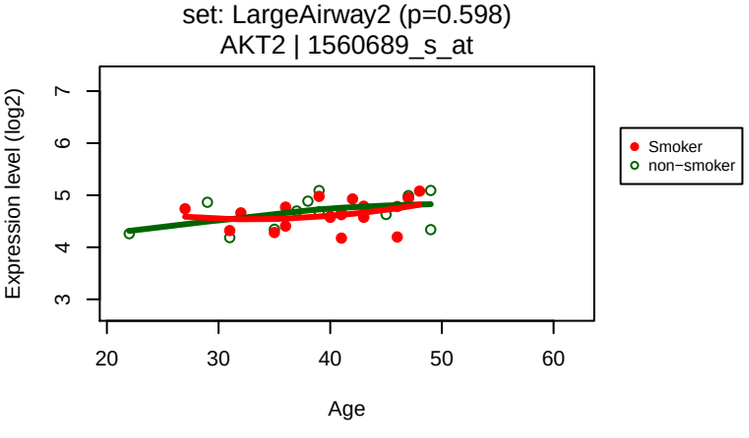

Gene expression ratio (smokers/nonsmokers)
gene: AKT2 | 1560689_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0575
p=0.1361
p=0.8849
p=0.1987
p=0.598
p=0.5322
p=0.9562
p=0.481
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

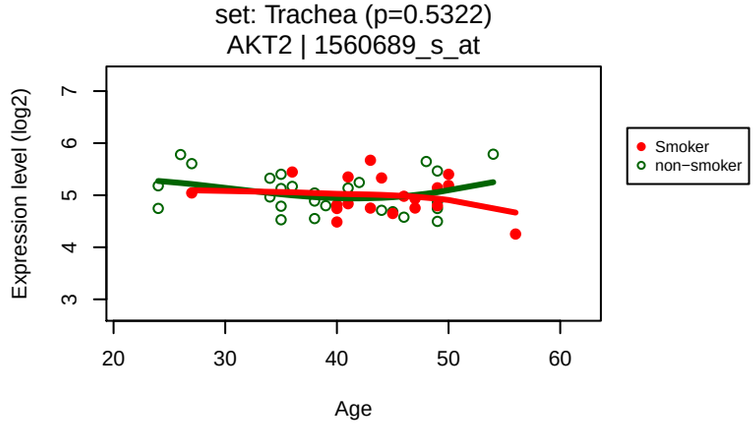

Gene expression ratio in Males
gene: AKT2 | 1560689_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1077
p=0.4948
p=0.7544
p=0.424
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT2 | 1560689_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8007
p=0.5694
p=0.8242
p=0.9804
SmallAirway2
LargeAirway2
Trachea
Macrophage2

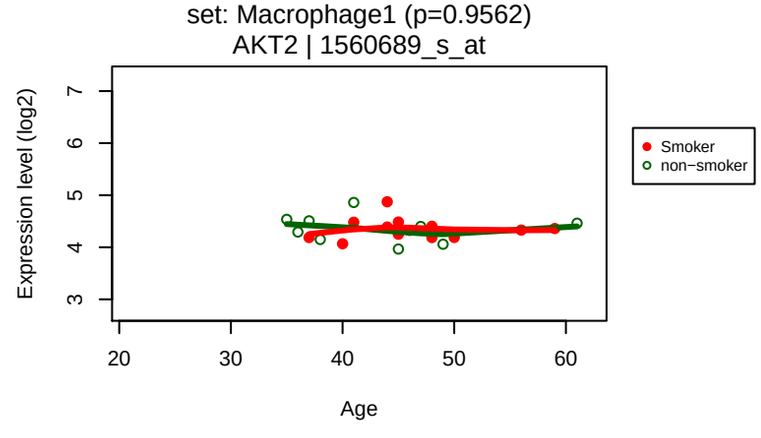

set: SmallAirway1 (p=0.0575)
AKT2 | 1560689_s_at
Expression level (log2)
Age
Smoker
non−smoker

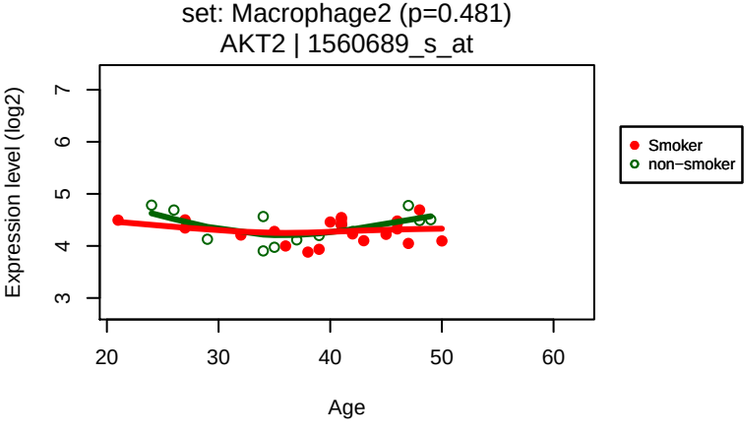

set: SmallAirway2 (p=0.1361)
AKT2 | 1560689_s_at
Expression level (log2)
Age
Smoker
non−smoker

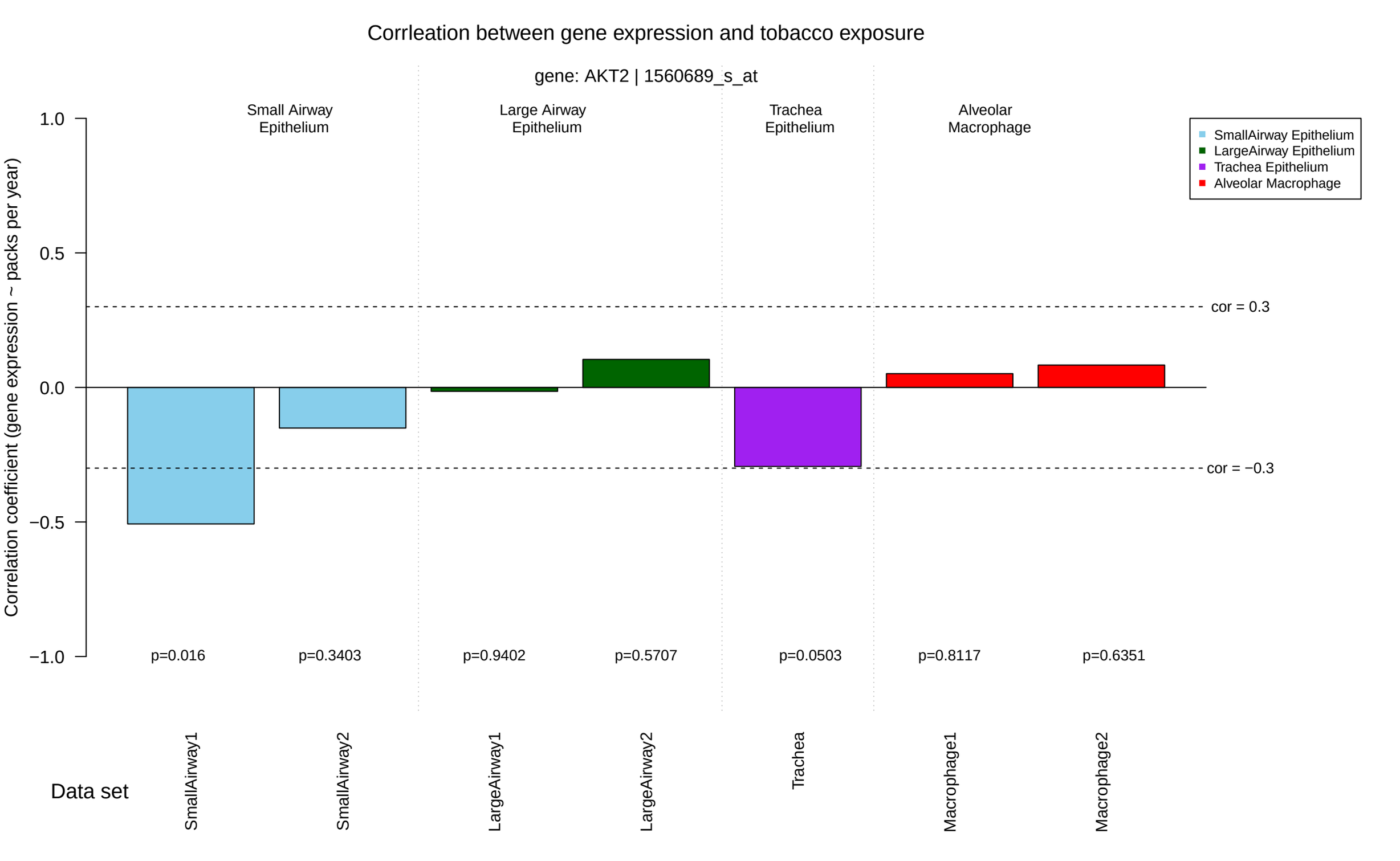

set: LargeAirway1 (p=0.1987)
AKT2 | 1560689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.598)
AKT2 | 1560689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5322)
AKT2 | 1560689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9562)
AKT2 | 1560689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.481)
AKT2 | 1560689_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT2 | 1560689_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.016
p=0.3403
p=0.9402
p=0.5707
p=0.0503
p=0.8117
p=0.6351
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2